

Variance

Variance measures how spread out the data points are from the average. If the data points are close to the mean, the variance is small.

Conversely, if they deviate significantly from the mean, the variance is large

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

For instance, if all students in a class have similar heights, the variance will be low. However, if the heights vary significantly, the variance will be high.